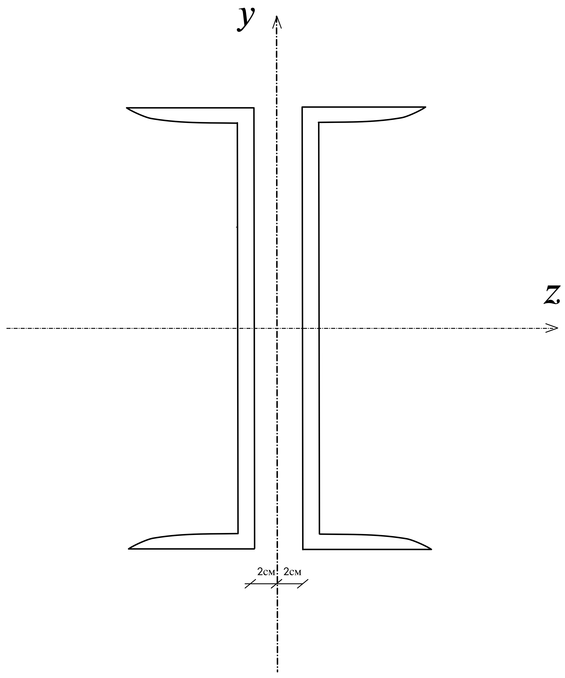



$\mu = 0{,}5$

$l = 5$

Решение

Условие устойчивости $\sigma = \frac{N}{\phi A} \le R\gamma_c$ ,отсюда $A \ge \frac{N}{\phi R}$

Первая попытка $\phi_1 = 0{,}7$ $A \ge \frac{340}{0{,}7 * 24} = 20{,}24\text{см}^2$

Выбираем 2 швеллера №10 $A = 10{,}9 * 2 = 21{,}8\text{см}^2$

Для одного швеллера по таблице сортамента $I_z = 174\text{см}^4$ $I_y = 20{,}4\text{см}^2$ $z_0 = 1{,}44\text{см}$

Для составного сечения

$I_z = (I_{z1} + a_y^2 * A_1) = 2I_z = 2 * 174 = 348\text{см}^4$

$I_y = (I_{y1} + a_z^2 * A_1) = 2\left(20{,}4 + (2 + 1{,}44)^2 * 10{,}9\right) = 298{,}77\text{см}^4$

$i_{min} = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{298{,}77}{21{,}8}} = 3{,}7\text{см}$

$\lambda = \frac{0{,}5 * 500}{3{,}7} = 67{,}57$

Из таблицы приложения 4 находим : $\phi = 0{,}766$

Проверяем условие устойчивости

$\sigma = \frac{340}{0{,}766 * 21{,}8} = 20{,}36\ \frac{\text{кН}}{\text{см}^2} < 24\frac{\text{кН}}{\text{см}^2}$

Определяем величину критической силы

$\lambda = 67{,}57 < \lambda = 100$ критическую силу определяем по формуле Ф. С. Ясинского :

$a = 31\left(\frac{\text{кН}}{\text{см}^2}\right)$ $b = 0{,}114\left(\frac{\text{кН}}{\text{см}^2}\right)$ $c = 0$

$\sigma_{cr} = a - b\lambda + c\lambda^2 = 31 - 0{,}114 * 67{,}57 = 23{,}29\left(\frac{\text{кН}}{\text{см}^2}\right)$

$F_{cr} = \sigma_{cr} A = 23{,}29 * 21{,}8 = 507{,}72\text{кН}$

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